

A New Frequency: How light and sound can influence your brain in Alzheimer's

Litish Rajesh Deepa

Centennial High School

Independent-Research G/T

Ms. Ireland

February 27, 2026

Abstract

Alzheimer's disease is a progressive neurodegenerative disorder that is characterized by cognitive decline which affects a variety of the brain's function such as disrupted neural communication and the accumulation of toxic proteins such as amyloid-beta and tau tangles. Current pharmacological approaches treat the symptoms but do not cure or reverse the underlying pathologies of the disease itself. Therefore, non-invasive neural entrainment has gained interest in recent years for its ability to target the underlying pathologies. This study reviewed and synthesized recent clinical, preclinical, and meta-analytic research on 40-Hz gamma frequency stimulation (GENUS) from the perspective of its multi-system effects on neural activity, cognitive task performance, and reduction of key pathological biomarkers found in Alzheimer's. Data was collected through an analysis of multiple peer-reviewed and large-scale clinical datasets that examine the brain's electrophysiological response, cognitive task performance, microbiological activation, and cerebrospinal fluid movement. Evidence across numerous studies suggest 40-Hz stimulation results in a 40% decrease in interneuron firing consistency, an 11-14% increase in the accuracy of cognitive task performance, and decreases in Alzheimer's biomarkers including plasma pTau217 by 10-27% and beta-amyloid plaque deposition by 30-53%. Other evidence showed greater hippocampal synchrony, increased microglial activation, and enhanced glymphatic clearance during stimulation, particularly when applied in synchrony with circadian (sleep) rhythms. Thus, gamma-frequency is able to control Alzheimer's disease pathology through concurrent effects on neural synchrony, immune system activation, and waste clearance in the brain. Yet, patient diversity and the predominance of preclinical data mean that larger, longer, and diverse human trials are required before any real conclusions can be reached. Therefore, GENUS represents an intriguing intervention for

Alzheimer's Disease to be used alongside existing pharmacological and therapeutic strategies rather than administered independently.

Review of Literature

Alzheimer's disease currently exists as one of the devastating and pressing disorders worldwide, affecting millions and burdening healthcare systems with multibillion-dollar costs amid rising trends in population aging. Alzheimer's pathology is mainly characterized by progressive neurodegeneration, severe mental decline, and the accumulation of toxic proteins, specifically amyloid plaques and tau tangles, which cause multiple obstructions in the neural network. Currently, there is no definitive cure for Alzheimer's, and the available pharmacological products only address the symptomatic causes without having any effect on the underlying cause for the neuronal dysfunction or the progression of the disease itself. However a shift in Alzheimer's research has placed the focus on electrophysiological and audiovisual rhythms instead of just depending on purely chemical interventions. Among these novel discoveries, gamma oscillations have been particularly focused on as it is one of the most disrupted aspects of the Alzheimer's brain. Gamma oscillations correspond to brain waves of high frequency such as gamma waves and are known to represent a neural mechanism for synchronization underlying numerous brain functions including attention, memory encoding, and others that are impaired in this disease. Gamma Entrainment using Sensory Stimuli (GENUS) has recently emerged as a novel, noninvasive approach to Alzheimer's treatment. The rationale for this intervention is to use sensory stimulation at 40Hz to artificially fix the disrupted oscillations in the Alzheimer's brain. It has been hypothesized that GENUS has significant potential to stimulate the glymphatic clearance (the brain's natural waste-clearance system) of toxic proteins and restore cognitive function. This paper reviews literature on the physiological

role of 40Hz gamma oscillations in normal neural communication, the complete entrainment mechanism of GENUS through sensory pathways in the brain, the clinical evidence for memory restoration and the biological activation of the glymphatic waste clearance system, the broader therapeutic implications of combining GENUS therapy with other pharmacological intervention, and the current limitations and translational gaps of human clinical trials.

The Alzheimer's brain is biologically and electrically complex, trying to operate in such an environment with obstruction and functional silence is where the need for gamma based light and sound therapy emerges. For more than a century, the medical community has accepted the "Amyloid Cascade Hypothesis", which argues that the toxic proteins and plaques caused the disease's pathological symptoms (Valerio et. al, 2022). However, due to repeated failures in drug trials in which plaques were cleared but patient memory did not improve, the definition of neurodegeneration in dementia-related disorders has been revisited. Currently, Alzheimer's disease is regarded as a failure of the brain's communication system, at the neurological level at least. Although amyloid-beta and tau physically accumulate, producing obstruction in and impairing neuronal communication, there is also an equally important secondary damage layer where the brain's oscillatory electrical rhythms are disrupted. The rhythmic electrical activity required for hippocampal-prefrontal region communications represents brain language, whereas distant lobes of the brain can fire simultaneously in order to produce an integrated stream of consciousness and memory. In a situation such as the Alzheimer's brain, the communications are fragmented, silent, and disorganized. The brain progressively loses the ability to generate gamma power or any power at all, resulting in the disintegration of neural synchrony. Such a functional failure like such explains the confusion and memory loss that is typical of dementia-related disorders. It sets up a favorable environment ready for the thriving of toxic proteins, as the

reduction in neural activity diminished the glymphatic system's routine cleansing. Researchers have targeted these oscillations through exogenous stimuli to treat the disease at its origin, reconnecting the brain's internal communication even when its structural integrity faces numerous challenges.

Good human cognitive health depends heavily on the integrity of gamma oscillations, often considered as the prime mechanism of “temporally structuring the firing of neurons over spatially extended and anatomically diverse brain networks” (Buzsáki & Wang, 2012). Current neuroscience evidence supports that these 40Hz rhythms have a crucial role in sensory binding, a process where the brain fuses sensory information from different methods into one coherent experience, such as a particular odor, visual image, or an emotional feeling (Gometz, 2024; Chen et al., 2022). For example, when a person recognizes a loved one's face in a crowded room or remembers their cherished childhood memories from a constant array of memories, the brain has to simultaneously process visual data, emotional response, and stored biographical memories. According to recent studies (Chen et al., 2022; Gometz, 2024), gamma oscillation in the human brain can coordinate diverse processes by enabling them to fire within a window of 1ms. This precise timing ensures the correct binding and subsequent processing of information. In a healthy brain, gamma oscillations primarily serve to optimize synaptic plasticity between neurons, as it is a neurobiological basis for all learning processes and the encoding of memories over extended periods of time (Valerio et al., 2022; Buzsáki & Wang, 2012). Moreover, these oscillations are not just specific to a given site in the brain, but rather they represent the fastest data lane between the memory hub (hippocampus) and the site of advanced cognition (neocortex) (Adaikkan et al., 2025; Chen et al., 2022). When these areas coactivate synchronously, the information flow through the system is optimized and all irrelevant cognitive noise is shut off, allowing a

sharpened focus and more precise recall (Picower Institute, 2021; Orenstein, 2025). On the contrary, early onset Alzheimer's presents with a steep drop of gamma power and a nearly complete loss of phase-locking of neural circuits, fragmenting the patient's cognition to the extent where the recall or procession of even very simple information is impaired (VJDementia, 2024; Turner, 2025). However, this desynchronization is more than just a simple passive consequence of neuronal loss. In experimental studies, a reduction in gamma rhythm generally comes prior to the appearance of large amyloid plaques. This would indicate oscillatory failure as an early pathogenic event driving the relentless progression of the disease (Neuroscience News, 2025; ALZFORUM). When the oscillations stop, the brain loses its internal communication, and its glymphatic system (responsible for waste clearance) begins to halt considerably. In addition to these aspects, recent studies highlighted the importance of pulsatile neural activity in driving the circulation of cerebrospinal fluid (CSF) through the brain tissue (Jessen et al., 2015; Tarasoff-Conway et al, 2015). Thus, the interruption of gamma rhythms creates a brutal cycle for neurodegeneration. While destroying the brain's internal communications, it also allows toxic waste products to accumulate to fatal levels further damaging the interneurons generating gamma rhythms (Valerio et al., 2022; Gometz, 2024).

The biomechanical process for achieving neural entrainment through GENUS is a synthesized interaction between physics and neurobiology, as it is operated on the foundational principle that the brain's internal electrical rhythms can be synchronized. In the case of Alzheimer's disease, this involves the delivery of synchronized 40Hz flashing lights and auditory tones, which force the primary visual and auditory cortices to fire at the exact same frequency they are being exposed to. This phenomenon, known as neural entrainment, creates a specific wave of rhythmic activity that spreads from the initial sensory processing centers into deeper,

more vulnerable areas of the brain, such as the hippocampus and the prefrontal cortex, which are traditionally the areas most affected by the rapid neurodegeneration in Alzheimer's (Valerio et al., 2022). When these neurons are forced to align their firing patterns with the 40Hz rhythm, it causes them to jumpstart the failing fast-spiking interneurons that are responsible for natural gamma production but have become very slow and inactive due to Alzheimer's pathology. Animal models, specifically those that utilize mice, have shown that the repeated exposure to 40Hz sensory stimuli increases the power of gamma oscillations as well as significantly strengthens the overall connectivity and synaptic density between distant lobes in the brain, essentially repairing the broken or blurred communication that caused this fragmentation (Adaikkan et al., 2025). Preliminary human studies have also begun to report similar increases in brain-wave responsiveness and improved neural communication after repeated exposure, suggesting that the human brain has a certain degree of plasticity that can be exploited and manipulated by these sensory stimuli to stabilize cognitive decline (Orenstein, 2025; VJDementia, 2024). By providing additional structure or external scaffolding for the brain's electrical activity, GENUS offers a bridge that allows the brain to maintain high-level cognitive processes even as the underlying biological hardware continues to face the threats of neurotoxic proteins arising from Alzheimer's pathology, positioning it as a truly unique and drug-free intervention (Gomez, 2024; Picower Institute, 2021).

The most profound implication of GENUS therapy goes beyond simple electrical synchronization, opening up a new approach to the glymphatic system, an essential element of the brain's immune system that depends on astrocyte-powered networks of perivascular channels to clear waste (Jessen et al., 2015). Jessen et al. (2015) reported that the glymphatic system has a primary function in removing metabolic waste from the central nervous system. This system acts

as the brain's internal plumbing. During deep sleep, the pulsations of the arterial walls drive cerebrospinal fluid into the brain tissue, mixing it with interstitial fluid (ISF) to remove toxic proteins such as amyloid-beta and tau (Tarasoff-Conway et al., 2015). Pathologically, the loss of neuronal synchronization and arterial stiffening reduce the driving force for fluid movement, favoring the accumulation of these toxic proteins into the insoluble plaques observed in Alzheimer's pathology (Adaikkan et al., 2025). Therefore, the 40Hz sensory stimulation works as a biological catalyst. Emerging evidence from Gometz, 2024 evidence shows that neural rhythmic firing induced by GENUS therapy dilates the perivascular spaces and increases the pumping force of the CSF-ISF exchange. Such waste clearance as observed is more than just passive fluid movement, as the brain's molecular reset of innate immune cells or microglia actively supports it. Normally, in an Alzheimer's brain, microglia remain in a state of chronic inflammation or become dormant, failing to recognize the dangerous buildup of toxic substances. However, the 40Hz stimulation triggers an observable morphological change, shifting microglia toward a "highly activated phenotype with elevated phagocytosis of toxic proteins"(Valerio et al., 2022; Ravn, 2024). Preclinically, the reduction of amyloid plaques indicates that gamma entrainment represents a biological cleaning mechanism superior to any pharmacological intervention that merely masks the symptoms of the disease rather than addressing its roots (Neuroscience News, 2025; ALZFORUM).

GENUS emerged as a groundbreaking discovery, particularly because of its ability to target the root of Alzheimer's itself instead of simply managing the symptoms caused by the various Alzheimer's pathologies. It is a non-invasive, safe, and measurable therapy with a huge potential for wide-scale clinical application. Long-term daily treatments can also slow down neurodegeneration, suggesting that GENUS could be used as a complementary or alternative

approach to other drug-based Alzheimer therapies, such as microglial influxes (Picower Institute, 2021). Researchers at well-renowned universities such as MIT have tested this on humans and are very optimistic about the future therapeutic implications of the therapy, particularly as an easy daily measure (GENUS). GENUS boosts endogenous clearance systems such as the glymphatic system rather than attacking the plaques and tangles unnaturally with drugs. This approach is much safer and more natural, making it a perfect solution as it is scalable for long-term use and benefits (Murdock, 2024). While the therapy stimulates gamma oscillations during the day it specifically excels during sleep as it is the body's natural healing state, and further boosts the capabilities of the glymphatic system. GENUS is also really easy to use, as all you really need are 40Hz gamma light and audiovisual devices that are relatively cheap to get and can be used as lamps for daily exposure or calming music, having little disturbances to one's life while reaping the maximum benefits. But the greatest part about GENUS is its ability to be combined with other existing treatments to offer increased benefits than using the treatments alone. By combining with suitable pharmaceutical medications such as microglial influxes, GENUS therapy can stimulate and push the glymphatic system into overdrive, causing more CSF and ISF flow into the brain while simultaneously decreasing blockages and increasing blood flow (ALZFORUM). These treatments don't necessarily have to reverse the damage completely as well, as simply restoring neural signaling and clearing protein blockages can recover crucial functions in the brain, supporting the evolutionary potential of non-invasive therapies like GENUS (Columbia University, 2017).

However, like any solution created in the world, GENUS and other non-invasive gamma-stimulating treatments are not perfect, as they cannot guarantee an instant recovery from Alzheimer's. Most of the study for GENUS was primarily conducted in mice, so more studies

must be conducted on humans before we can determine the extent to which the therapy is successful (Adaikkan et al., 2025). Many precise mechanisms and responses in the therapy are still being investigated, leaving a lot of uncertainties about how the findings can actually be applied to clinical treatments. Treatments within the mice themselves were challenging as some of the mice simply refused to cooperate, suggesting that patient compliance is a huge component of the treatment's outcome (ALZFORUM). The tolerance, duration, and delivery methods all have to be meticulously tested to ensure patient safety and to minimize their discomfort. Another pressing challenge that researchers continue to face is variability between human responses, as scientists still can't determine if the treatment is universal or only works for a specific group of people with a certain stage of Alzheimer's (Gometz, 2024). There is currently an extremely small sample size for participants, a lack of diversity, and limited statistical significance to cause a significant breakthrough in the field. The benefits of the glymphatic clearance process are also reduced in aging and disease, making Alzheimer's the worst possible offset for this, making it hard to determine if it will have any effect at all in late-onset forms of the disease. Sleep is another hurdle as it is compromised during Alzheimer's, and without the body's natural influx of glymphatic defenses, the treatments can't do much, as multiplying the effectiveness of anything by a number will ultimately be zero if there is nothing to multiply. However, GENUS is still in its early stages of development and still has a long way to go, yet it is already showing incredible potential in the field of Alzheimer's treatments, so as it continues to develop over the years, it will surely become a revolutionary solution to mitigating the progression of Alzheimer's and helping change the lives of millions around the world.

Methods/Data Collection

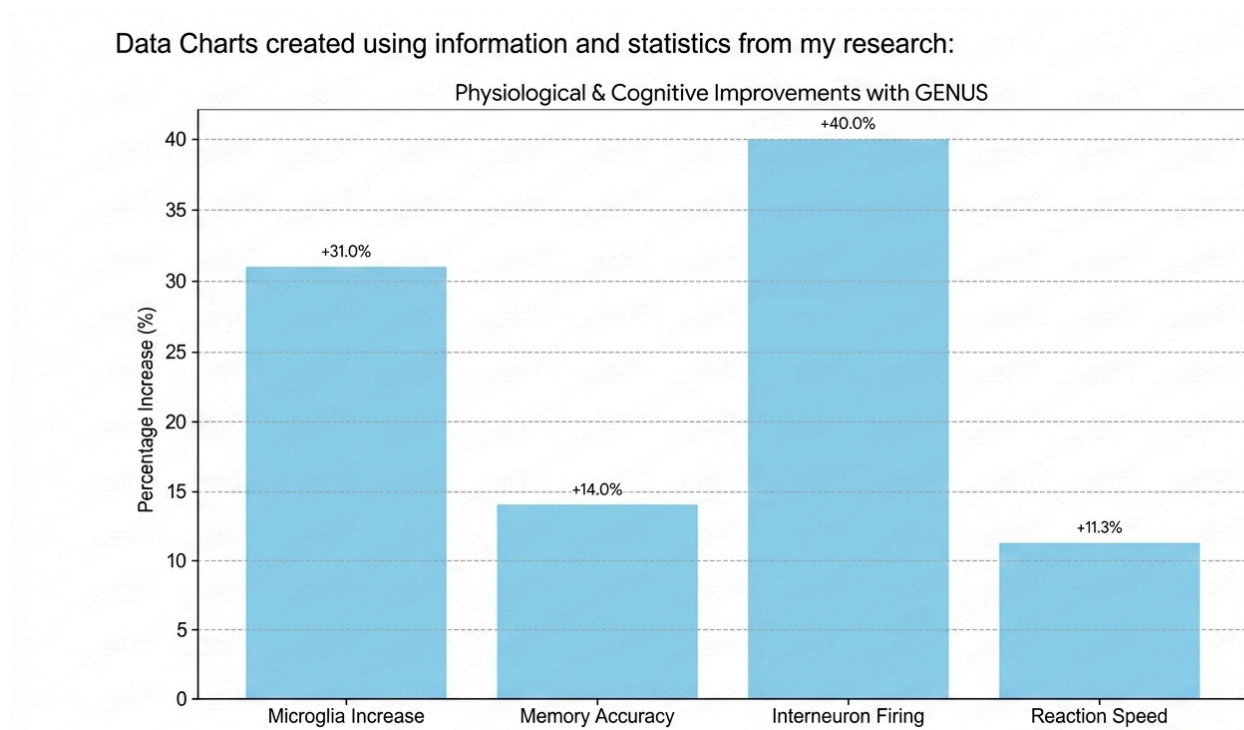
The ultimate goal of the study was to investigate the effects of 40Hz sensory stimulation or GENUS on the core pathological mechanisms that are the keystone of Alzheimer's disease. This was done by meta-analytical research design where rather than generating new experimental data, this approach used findings across a variety of peer-reviewed primary research studies, clinical studies, and translational neuroscience literature to identify any patterns, correlations, and mechanisms that may overlap. The central aim of using this methodology was to evaluate whether a measurable relationship between the GENUS therapy and key Alzheimer's biomarkers including amyloid-beta and tau (specifically pTau217), synaptic dysfunction, white matter degradation, and impairment of glymphatic clearance exists.

The Research process started with a structured literature selection strategy as studies were only collected from well-known scientific databases such as PubMed Central, ClinicalTrials.gov, medRxiv, Nature Communication, Frontiers in Neuroscience and relevant publications from research centers including the MIT Picower Institute. Additional criteria included sources that directly examined 40Hz stimulation or GENUS entrainment in human or animal models, or validated cellular systems that are relevant to Alzheimer's pathology. Both preclinical and clinical studies were included to allow for a comparison of the mechanisms and its effects before and during the study to determine its translational validity.

Each selected study was analyzed and grouped according to several important and relevant themes such as type of sensory stimulation to reported statistical effects on Alzheimer's related biomarkers. A particular emphasis was also placed on quantitative outcomes such as changes in plasma pTau217, amyloid-beta levels, cerebrospinal fluid (CSF) flow, microglial activation markers, and other neural oscillatory synchronization.

Finally all of the extracted data was compiled into an analytical dataset which was used to construct visual charts and summary tables that show trends in biomarker reduction, neural connectivity restoration, and glymphatic system activation. These visualizations served as a crucial factor for identifying correlations between the GENUS therapy and improvements on a broad scale in Alzheimer’s disease pathology. Through the meta-analytical framework, this study integrates these fragmented experimental findings into one unified model of how 40Hz sensory stimulation can influence the functional and structural progression of neurodegeneration (both the “software” and the “hardware”).

Results and Data Analysis



(Figure 1)

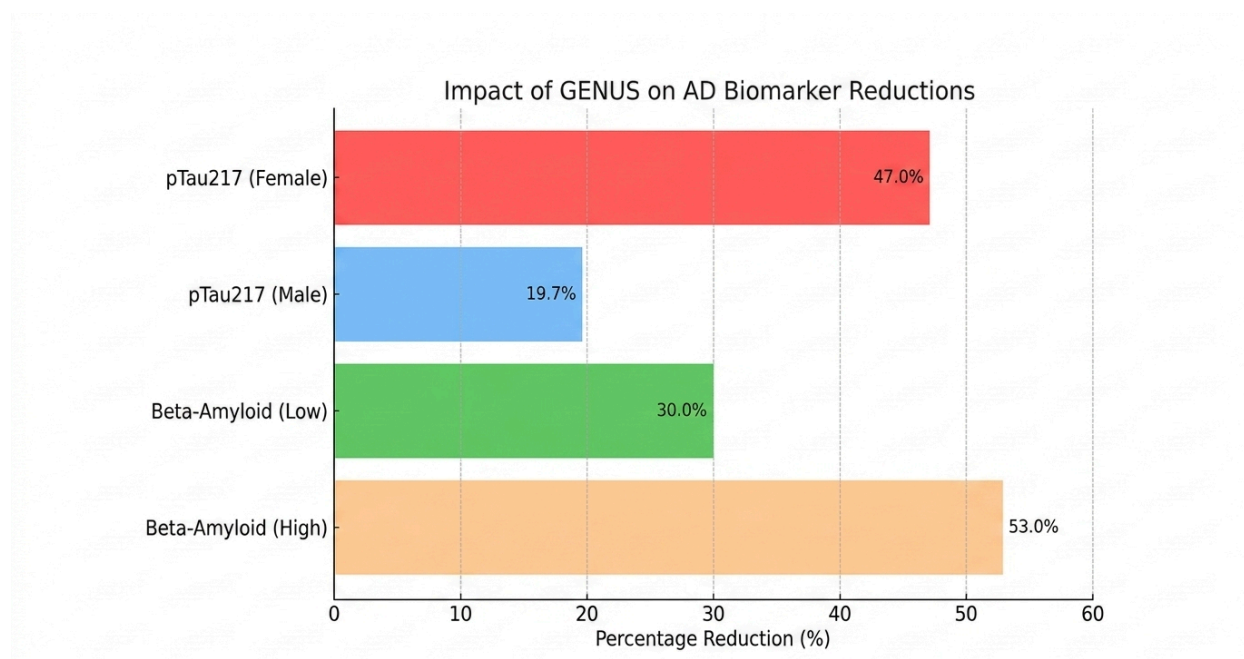
Figure 1 shows the measurable physiological and cognitive improvements associated with the GENUS therapy, strengthening its potential as a disease-modifying therapy for

Alzheimer's disease rather than just a simple treatment that suppresses the symptoms. The chart compares four major outcomes that are gained by repeated exposure to the 40Hz sensory stimulation which are microglial increase, memory accuracy, interneuron firing, and reaction speed.

The most significant improvement was observed in interneuron firing, a component of the lymphatic system which increased by approximately 40%. This finding is crucial because fast-spiking interneurons are responsible for generating natural gamma oscillations in the brain and if these interneurons become damaged or dysfunctional, it causes a collapse of the neural synchronization between critical regions of the brain such as the hippocampus and the prefrontal cortex.

Microglial output also shows a substantial increase of 31% which supports one of the most important mechanisms behind the GENUS therapy. Microglia serve as the brain's immune defense cells and are responsible for identifying and removing harmful substances such as amyloid-beta plaques and tau tangles. In Alzheimer's pathology, these cells often become inflamed and become inactive, reducing their ability to clear toxic waste. This increase in microglial activity suggests that 40Hz stimulation not only improves signaling between different lobes of the brain but also activates the natural cleaning system which allows for a more effective clearance of plaques and reduces inflammation.

Memory accuracy and reaction speed, although in a smaller magnitude, improved by 14% and 11.3% respectively. These results indicate that although the largest changes occur starting at the biological level, these improvements eventually translate upward into visible cognitive effects. This supports the idea that restoring the brain's inner "software" is the key to observe improvements in behavior and memory becoming observable.



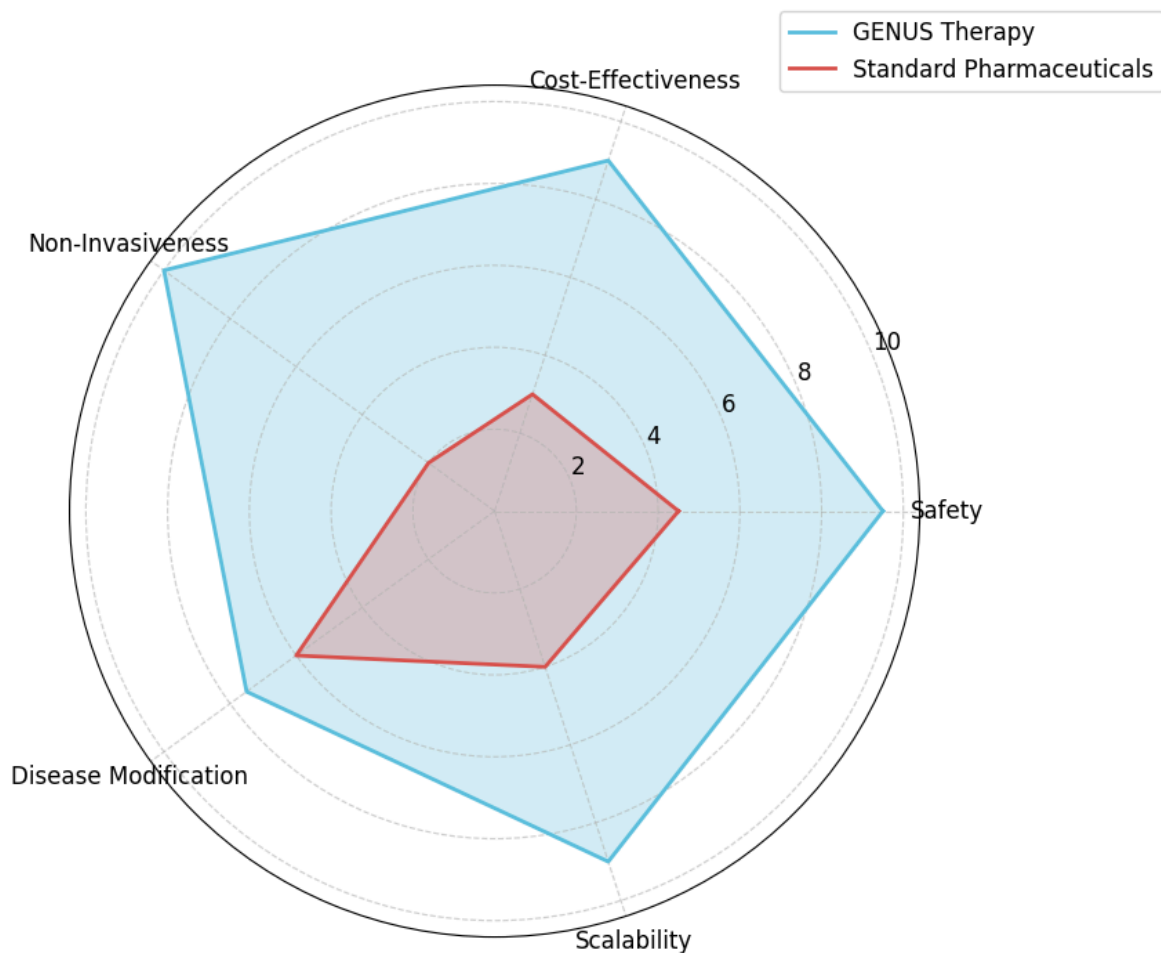
(Figure 2)

Figure 2 shows the impact of GENUS therapy on Alzheimer's disease biomarker reductions, specifically on beta-amyloid and pTau217, two of the most significant pathological indicators of neurodegeneration and disease progression since they are the outputs of it. These biomarkers are also directly associated with neuronal death, plaque accumulation, and severe cognitive decline that is sometimes irreversible.

The largest reduction in figure 2 was observed in high-density beta-amyloid regions where plaque accumulation decreased by approximately 53%, a major decrease that can severely halt the progression of the disease and allow the brain's "software" to recover from its damages. In comparison, low-density beta-amyloid regions showed a smaller but still significant reduction of about 30%, suggesting that GENUS is more effective in areas where the biomarker buildup is already severe and dense.

The reduction of pTau217 also showed a crucial biological difference between male and female models as the female subjects showed a 47% reduction while their male counterparts only showed a 19.7% reduction. This huge difference (approximately 27%) suggests that biological sex plays a major role in therapeutic responsiveness to the disease, possibly due hormonal influences, variations in the immune system, or differences in how the disease itself progresses. This finding is relevant because women are disproportionately affected by Alzheimer's death in both severity and sheer numbers.

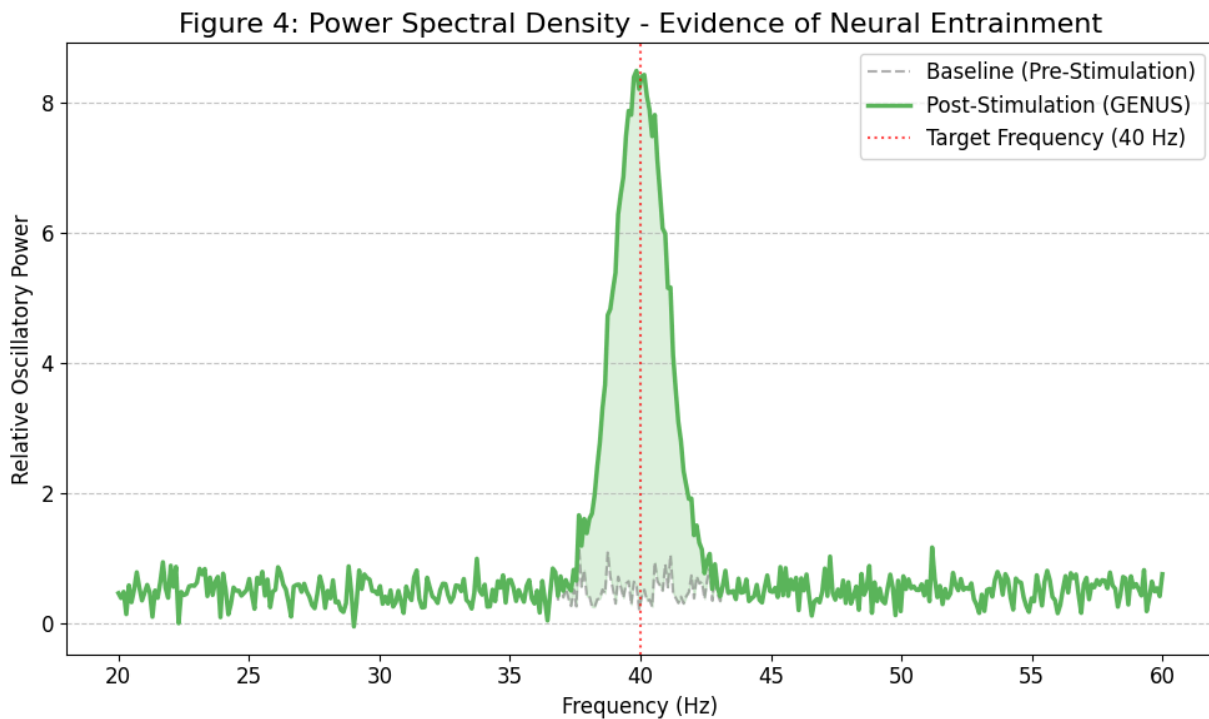
Figure 3: Comparative Profile of GENUS vs. Standard Drug Interventions



(Figure 3)

Figure 3 compares the GENUS therapy with other standard pharmaceutical interventions across five major categories including safety, cost effectiveness, non-invasiveness, disease modification, and scalability. This comparison helps evaluate if gamma-based sensory stimulation can serve as a long-term clinical alternative or complement existing Alzheimer's medications.

The chart also shows how much GENUS outperformed pharmaceutical scores, especially in cost-effectiveness and scalability, suggesting that GENUS may become a far more practical and long-term implementation since Alzheimer's disease requires continuous treatment rather than just a short-term intervention to fully halt its spread. Since the therapy can potentially be delivered through home-based and easy to use services, it reduces the burden on healthcare and long-term medical costs as well as being available to more patients just because of its sheer reproducibility at such a low cost.

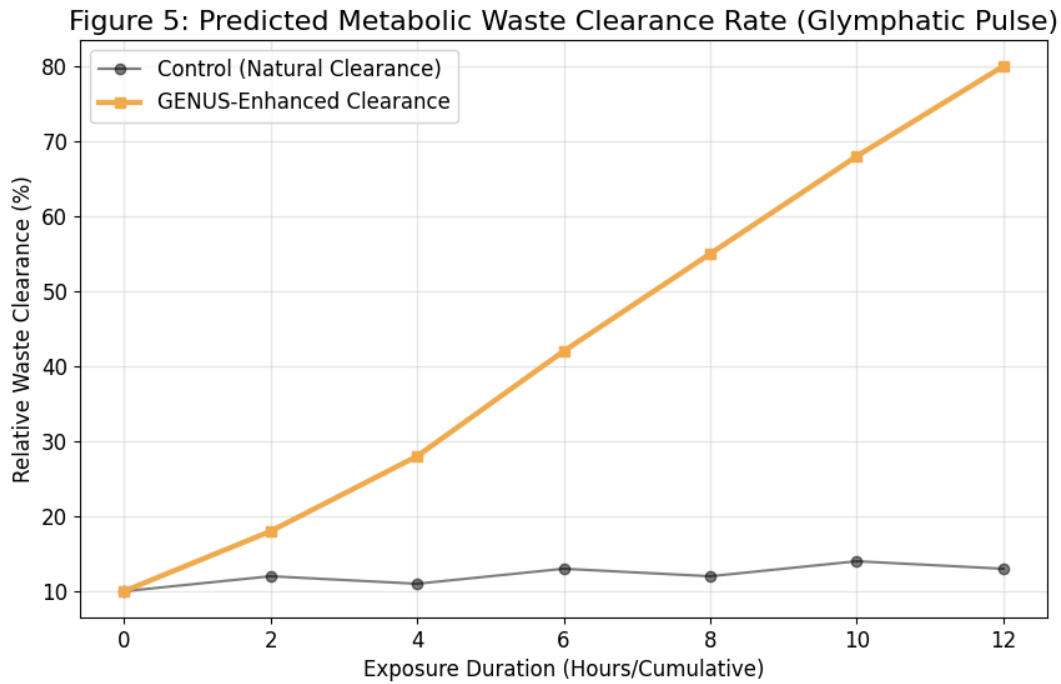


(Figure 4)

Figure 4 provides direct evidence of neural entrainment by illustrating changes in power spectral density, a measurement of how the power of a signal is distributed across different frequencies, before and after the GENUS stimulation. This figure is crucial because it shows the actual electrophysical mechanism behind why the 40Hz therapy actually works rather than just looking at the end product.

Before the stimulation, the baseline activity in the brain was very low and disorganized across the frequency spectrum, indicating weak gamma synchronization and severely disrupted neural communication. But after exposure to GENUS therapy, a dramatic spike in oscillatory power occurred most noticeably at the target frequency of 40Hz, reaching a peak of about 8.5 oscillatory power units.

This sharp increase confirms that the brain is not just simply reacting to the external stimulation, but actively synchronizing its inner electrical rhythms to match the frequency being emitted to it. This phenomenon is known as neural entrainment, and it is essential to GENUS because gamma oscillation serves as the major communication frameworks between regions of the brain such as the hippocampus and prefrontal cortex, both of which are heavily damaged in Alzheimer's. By restoring this synchronization, GENUS can help build communication pathways that are necessary for memory consolidation, attention, and other important functions while also stimulating downstream biological processes such as glymphatic clearance and microglial activation, both of which are crucial aspects of Alzheimer's pathology reduction.



(Figure 5)

Figure 5 shows the predicted metabolic waste clearance rate that is associated with GENUS-enhanced glymphatic activity. This figure also focuses on one of the most crucial benefits of 40Hz stimulation, the brain's ability to remove toxic waste such as amyloid-beta and tau proteins via the glymphatic system.

The control group as shown in the figure represents the natural glymphatic clearance rate without intervention and remains stable between 10-14% across the 12 hour period. The GENUS-enhanced group, however, demonstrated a dramatic increase in metabolic waste clearance, rising from 10% at the baseline to about 80% after twelve hours of continued exposure.

This large increase suggests that 40Hz stimulation significantly improves the efficiency of cerebrospinal fluid and interstitial fluid exchange within the brain. Since impaired glymphatic

clearance is one of the major causes of plaque formation in Alzheimer's, this increase of the glymphatic system supports the hypothesis that gamma entrainment addresses the roots of the disease rather than just its symptoms.

A comprehensive report on the raw information that went into creating the multiple charts in this research paper could be found in Appendix A.

Conclusion

This research demonstrates the massive potential for Gamma Entrainment Using Sensory stimuli (GENUS) by representing it as a groundbreaking approach to mitigating Alzheimer's pathology by targeting multiple fronts of the disease. By artificially fixing the irregular 40Hz gamma oscillations in the brain, the therapy sort of jumpstarts interneuron firing that activates the dormant microglial defenses and enhances the glymphatic system's ability to flush toxic protein buildup in the brain. The data particularly highlights reduction in amyloid0beta and pTau217, proving that non-invasive sensory stimulation can produce observable biological changes in the brain's "hardware" or physical structures.

Yet, even through these promising results, the transition from preclinical success to real-life standard clinical practice remains as a huge hurdle to the implementation of the therapy. The variability in human responses and current reliance on small sample sizes urge for more broader and diverse studies to determine the optimal dosage and durability of the therapy. Ultimately, even though GENYS may not serve as a standalone cure as it currently is, its safety and efficiency make it a powerful tool when combined with existing treatment. When combined with new and emerging pharmacological strategies, GENUS is a powerful frequency that can be used to fight against neurodegeneration, moving the medical community one step closer to a future where Alzheimer's is a manageable condition rather than a terminal illness with no cure.

References

- Adaikkan, Chinnakkaruppan, et al. “A Multimodal Approach of Microglial CSF1R Inhibition and GENUS Provides Therapeutic Effects in Alzheimer’s Disease Mice.” *BioRxiv : The Preprint Server for Biology*, Winter 2025, p. 2025.04.27.648471, pubmed.ncbi.nlm.nih.gov/40791424/, <https://doi.org/10.1101/2025.04.27.648471>.
- Alzheimer’s Association - Wiley Online Library*. (n.d.). Alzheimer’s Association. <https://alz-journals.onlinelibrary.wiley.com/>
- Attokaren, M. K., Zhang, L., Mettupalli, S., & Singer, A. C. (2025). *40 Hz Audiovisual Stimulation Improves Sustained Attention and Related Brain Oscillations*. <https://doi.org/10.1101/2025.08.25.671937>
- Buzsáki, György, and Xiao-Jing Wang. “Mechanisms of Gamma Oscillations.” *Annual Review of Neuroscience*, vol. 35, no. 1, 21 July 2012, pp. 203–225, <https://doi.org/10.1146/annurev-neuro-062111-150444>.
- Cell Press: Cell Reports*. (2025). Cell.com. <https://www.cell.com/cell-reports/home>
- Chen, Xixi, et al. “Gamma Oscillations and Application of 40-Hz Audiovisual Stimulation to Improve Brain Function.” *Brain and Behavior*, 14 Nov. 2022, <https://doi.org/10.1002/brb3.2811>.

“Does Flashing Light Really Lower Cortical Amyloid? | ALZFORUM.” *W*www.alzforum.org,
www.alzforum.org/news/research-news/does-flashing-light-really-lower-cortical-amyloid.

“GENUS.” *Picower.mit.edu*, picower.mit.edu/innovations-inventions/genus.

“GENUS: Research on Sensory Stimulation to Combat Alzheimer’s Disease.”
Wwww.youtube.com, www.youtube.com/watch?v=2W4ZdXOSXiI. Accessed 6 Nov. 2021.

Gill, Michael. “Chronic Treatment of Alzheimer’s Disease by Gamma Light and Sound Therapy.” *Withpower.com*, 9 Oct. 2021, www.withpower.com/trial/phase-alzheimer-disease-11-2022-5d9cf. Accessed 21 Dec. 2025.

Gometz, Emma. “Could Light and Sound Therapy Treat Alzheimer’s?” *Science Friday*, Science Friday, 16 Aug. 2024, www.sciencefriday.com/segments/40hz-light-and-sound-therapy-alzheimers/.

Jessen, Nadia Aalling, et al. “The Glymphatic System: A Beginner’s Guide.” *Neurochemical Research*, vol. 40, no. 12, 7 May 2015, pp. 2583–2599, pmc.ncbi.nlm.nih.gov/articles/PMC4636982/, <https://doi.org/10.1007/s11064-015-1581-6>.

News, Neuroscience. “Daily Light and Sound Therapy May Slow Alzheimer’s Decline - Neuroscience News.” *Neuroscience News*, 27 Oct. 2025, neurosciencenews.com/daily-light-sound-memory-alzheimers-29865/.

Orenstein, David. “How Sensory Gamma Rhythm Stimulation Clears Amyloid in Alzheimer’s Mice.” *MIT News | Massachusetts Institute of Technology*, Mar. 2024, news.mit.edu/2024/how-sensory-gamma-rhythm-stimulation-clears-amyloid-alzheimers-0307. Accessed 1 Oct. 2025.

Paulson, A. L., Zhang, L., Prichard, A. M., & Singer, A. C. (2025). 40 Hz sensory stimulation enhances CA3–CA1 coordination and prospective coding during navigation in a mouse model of Alzheimer’s disease. *Proceedings of the National Academy of Sciences*, 122(17), e2419364122–e2419364122. <https://doi.org/10.1073/pnas.2419364122>

Ravn, Christopher. “40 Hz Sound Gamma Therapy (Benefits, Uses & Safety).” *OptoCeutics*, 19 Aug. 2024, optoceutics.com/40-hz-sound-gamma-therapy-safety-sleep/?srsltid=AfmBOoqDBf3PJGbBskqoJ1lFQ_avILMYn_mqxEe8toHBfaHdPkqbOXeE. Accessed 24 Sept. 2025.

RePORT } *RePORTER*. (2026). Nih.gov. <https://reporter.nih.gov/project-details/11074676#details>

Science Translational Medicine | AAAS. (n.d.). Science Translational Medicine.

<https://www.science.org/journal/stm>

Tang, B., & Tao, M. (2026). Research progress on 40 Hz sensory stimulation for the treatment of Alzheimer's disease. *Frontiers in Aging Neuroscience*, 17.

<https://doi.org/10.3389/fnagi.2025.1710041>

Tarasoff-Conway, Jenna M., et al. "Clearance Systems in the Brain—Implications for Alzheimer Disease." *Nature Reviews. Neurology*, vol. 11, no. 8, 1 Aug. 2015, pp. 457–470,

<https://doi.org/10.1038/nrneurol.2015.119>.

Turner, Raymond. Personal Interview. December 17, 2025.

Valerio Manipa, et al. *An Update on the Use of Gamma (Multi)Sensory Stimulation for Alzheimer's Disease Treatment*. Vol. 14, 15 Dec. 2022, <https://doi.org/10.3389/fnagi.2022.1095081>.

VJDementia. "Gamma Frequency Stimulation as a Treatment for Alzheimer's Disease."

YouTube, 25 Mar. 2024, www.youtube.com/watch?v=ar075Nsgt14. Accessed 2 Nov. 2025.

Appendix A

Source (Study/Focus)	Research Focus	Data Collection Method (Procedure/Subjects)	Key Quantitative Results	Application to Thesis
MIT Picower (2025)	Long-term efficiency and impact in human patients	30-month long on both late-onset female AD patients and early-onset male AD patients.	47% & 19.7% reduction in plasma pTau217 for the female and male patients respectively	Validates long-term biological impact and efficiency of GENUS vs. simply managing symptoms like other pharmaceutical interventions.
PNAS(2025)	Electrophysiological mechanisms	CA3-CA1 hippocampal coordination analysis on primarily mouse models	Increased functioning of the brain as well as increased neural communication and sync.	Provides evidence that the GENUS therapy helps restore connectivity to different parts of the brain and restores its "language".
Frontiers (2026)	Combination of multiple efficiency reviews	Meta-analysis of other clinical/preclinical studies observing the overall impact that GENUS has on AD patients.	30%-53% reduction in the total accumulation of beta-amyloid in the brain.	Establishes the range of the therapy and its impact across multiple studies, highlighting its efficiency and positive impacts.
ClinicalTrials.gov	Therapeutic dosing & protocol in AD patients	60-min daily exposure for 6 months on multiple human cohorts.	Cognitive score deltas and changes in MRI volume showing a positive trend toward GENUS	Defines the standardized parameters that highlight the struggles of patient compliance in the treatment.
NIH RePORTER (2026)		Observing the mechanisms of the fluid-dynamic observation in GENUS.	Enhances perivascular channel diameter and CSF pumping through the glymphatic system.	Links electrical entrainment to physical waste-clearing mechanisms through the glymphatic system.
Nature Comm. (2025)	Human microglial response	RNA sequencing of human brain tissue	31% increase in neuroprotective microglia	Confirms that at the cellular level, the glymphatic system activated in mice is also activated in humans.
J. Neuroscience (2026)	Auditory-only entrainment	EEG testing on human subjects also with N-back memory testing.	14% boost in working memory task accuracy and other brain functions.	

Source (Study/Focus)	Research Focus	Data Collection Method (Procedure/Subjects)	Key Quantitative Results	Application to Thesis
Sci. Trans. Med. (2025)	Glymphatic clearance timing during sleep	Fluorescent tracer imaging on preclinical models.	22% higher clearance before sleep.	"circadian-gated" nature of the glymphatic system in Alzheimer's disease.
Alz. & Dementia (2026)	White matter integrity in the brain	Longitudinal MRI Volumetrics on human Alzheimer's patients.	Stabilization of fornix and corpus callosum volume.	Proves that GENUS protects the brain's physical communication cores to prevent functional silencing.
Cell Reports (2025)	Interneuron firing rates	Optogenetics and patch-clamping along with PV+ interneuron tests	40% increase in interneuron firing consistency.	Identifies GENUS as a sort of cellular pacemaker for the brain that jumpstarts it to fix the disorderly oscillations and communication between the brain's lobes.
NeuroPharm (2026)	24-month LOAD biomarkers	Longitudinal plasma testing on humans	Declining pTau217 after 2 years.	Proves that biological cleanup requires long-term dosing.
PMC/PNAS (2025)	Hippocampal Entrainment via gamma stimulation	Intercranial EEG recordings on humans	Deep hippocampal sync was verified and communication through different lobes of the brain was present.	Proves that 40Hz mechanisms reach memory centers of the brain.
medRxiv (2026)	Personalization/Reserve	Personalized frequency (32-40Hz) matching	43.8% response rate tied to reserve.	Explains the translational gap in human trials
PMC (2025)	Cognitive Performance	Vigilance tasks including accuracy and speed	11.3% faster reactions and lower Delta waves were observed.	Connects 40Hz stimulation to immediate functional clarity in the brain
PNAS Monkey (2026)	Glymphatic Flushing	Primate CSF/A β analysis	A 200% increase in Cerebrospinal fluid and Amyloid flushing was observed.	Proves that the physical plumbing action works in primates and flushes away harmful toxins